
Memory-Efficient Pseudo-Labelling for Online Source-Free Universal Domain Adaptation using a Gaussian Mixture Model

Winter Conference on Applications of Computer Vision (WACV) 2025

Presented By:

Jay Fiske

Under the supervision of:

Udaya Kumar

Under the guidance of:

Prof. C. Krishna Mohan



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Problem Statement

Online Source-Free Universal Domain Adaptation (SF-UniDA):

Goal: Adapt a pre-trained source model to an unlabeled target data stream.

Three Key Paradigms:

1. **Universal:** Handle Partial-Set (PDA), Open-Set (ODA), and Open-Partial-Set (OPDA) scenarios without prior knowledge of the category shift.
2. **Source-Free:** No access to source data during adaptation; only the pre-trained model is available.
3. **Online:** Target data arrives in batches $\{x_{i,k}^t\}$ and is accessed only once.

New Constraint: Memory-Efficiency. Saving knowledge from previous batches must be done with minimal memory footprint

Methodology

Core Objective: The goal is to achieve universal domain adaptation (DA) in an **online, source-free, and Memory efficient manner**.

Mathematical Framework:

- **Source Knowledge:** A neural network f is pre-trained on a source dataset $\mathcal{D}_s = \{\mathbf{x}_i^s \in \mathcal{X}_s, y_i^s \in \mathcal{Y}_s\}_{i=1}^{N_s}$
- **Target Data:** The model must adapt to unlabelled target data $\mathcal{D}_t = \{\mathbf{x}_i^t \in \mathcal{X}_t, ? \in \mathcal{Y}_t\}_{i=1}^{N_t}$
- **Model Architecture:** The network f is divided into a feature extractor g and a classifier h , expressed as $f = h \circ g$.

Methodology

1. GMM-Based Knowledge Transfer

Core Idea: Use a Gaussian Mixture Model (GMM) to capture the distribution of known classes in the feature space as a compact memory of past batches.

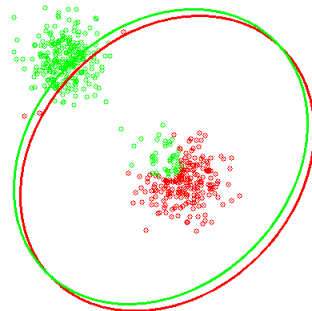
- **Iterative Updates:** Parameters (means μ_k and covariance Σ_k) are updated batch-wise via a weighted recalculation inspired by the EM algorithm.
- **Weighting Strategy:** Uses model predictions ($f(x)$) for intra-batch weighting and a class-wise sum of softmax outputs (s_k) for inter-batch weighting.
- **Decay Factor (α):** Controls the balance between retaining previous knowledge and adapting to new data.

$$s_k(c) = \alpha \cdot s_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) ,$$

Methodology

$$\hat{\boldsymbol{\mu}}_k(c) = \frac{\alpha \cdot s_{k-1}(c) \cdot \hat{\boldsymbol{\mu}}_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) \cdot r_{i,k}}{s_k(c)}$$

$$\hat{\boldsymbol{\Sigma}}_k(c) = \frac{\begin{pmatrix} \alpha \cdot s_{k-1}(c) \cdot \hat{\boldsymbol{\Sigma}}_{k-1}(c) + \sum_{i=1}^{N_b} f_c(\mathbf{x}_{i,k}^t) \\ \cdot (r_{i,k} - \hat{\boldsymbol{\mu}}_k(c)) (r_{i,k} - \hat{\boldsymbol{\mu}}_k(c))^T \end{pmatrix}}{s_k(c)}$$



Methodology

2. Pseudo-Labelling & OOD Detection

Strategy: Assign labels to known classes using the GMM while identifying "unknown" samples via Out-of-Distribution (OOD) detection.

- **Likelihood Estimation:** Calculates the probability of a sample belonging to each Gaussian distribution (known class).
- **Entropy-Based Rejection:** Uses the normalized Shannon entropy of GMM likelihoods to assess confidence.
- **Dual Thresholding:** Employs τ_k (known) and τ_u (unknown) to filter uncertain samples and prevent "negative transfer".

$$I(\mathbf{p}_{i,k}) = -\frac{1}{\log |\mathcal{Y}_s|} \cdot \mathbf{p}_{i,k}^T \cdot \log \mathbf{p}_{i,k}$$

$$\hat{y}_{i,k}^{\text{pl}} = \begin{cases} \arg \max_{c \in \mathcal{Y}_s} \mathbf{p}_{i,k} & I(\mathbf{p}_{i,k}) \leq \tau_k \\ |\mathcal{Y}_s| + 1 & I(\mathbf{p}_{i,k}) \geq \tau_u \\ \text{None} & \text{otherwise} \end{cases}$$

Methodology

3. Adaptation via Contrastive Loss

Contrastive loss is introduced which supports this hypothesis by enforcing the class-wise clustering of samples in the feature space.

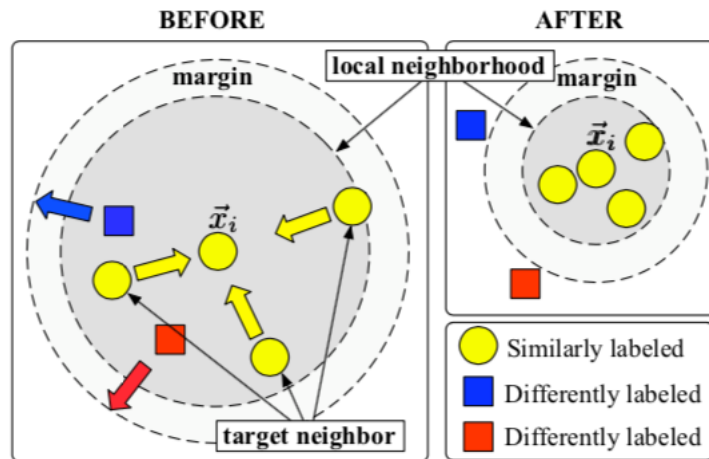
Function: Optimizes the feature extractor (g) to create a meaningful and well-separated feature space.

- **Positive Pairs:** Samples are pulled toward other samples of the same class and the corresponding GMM cluster center (mean).
- **Negative Pairs:** Distances are maximized between samples of different classes and between known samples and the "unknown" class.
- **Source-Free Advantage:** Uses GMM means as cluster centres instead of requiring source prototypes (like OWTTT and COMET-P)
- **Data Augmentation:** Each batch is doubled with augmented versions to ensure the feature space is robust to input variations.

Methodology

$$\mathcal{L}_C = - \sum_{j=1}^{2N_b} \sum_{i=1}^{2N_b} \mathbb{1}(\hat{y}_{j,k}^{\text{pl}} = \hat{y}_{i,k}^{\text{pl}}) \log \frac{\exp\left(\frac{\langle \tilde{r}_{j,k}, \tilde{r}_{i,k} \rangle}{\tau}\right)}{\sum_{l=1}^{2N_b} \exp\left(\frac{\langle \tilde{r}_{l,k}, \tilde{r}_{i,k} \rangle}{\tau}\right)}$$

$$- \sum_{c=1}^{|\mathcal{Y}_s|} \sum_{i=1}^{2N_b} \mathbb{1}(\hat{y}_{i,k}^{\text{pl}} = c) \log \frac{\exp\left(\frac{\langle \hat{\mu}_k(c), \tilde{r}_{i,k} \rangle}{\tau}\right)}{\sum_{l=1}^{2N_b} \exp\left(\frac{\langle \hat{\mu}_k(c), \tilde{r}_{l,k} \rangle}{\tau}\right)}$$



Methodology

4. Classifier Optimization via KL Divergence

Function: Trains the classifier (h) to handle feature representations that fall outside the original source training

- **Known Class Goal:** Maximizes the KL divergence between classifier output and a uniform distribution to encourage confident, "sharp" predictions.
- **Unknown Class Goal:** Minimizes the KL divergence for samples labelled as "unknown," forcing the output toward a uniform distribution (high entropy).
- **Joint Objective:** The final loss (L) combines the contrastive loss and KL divergence loss, weighted by a hyperparameter λ .

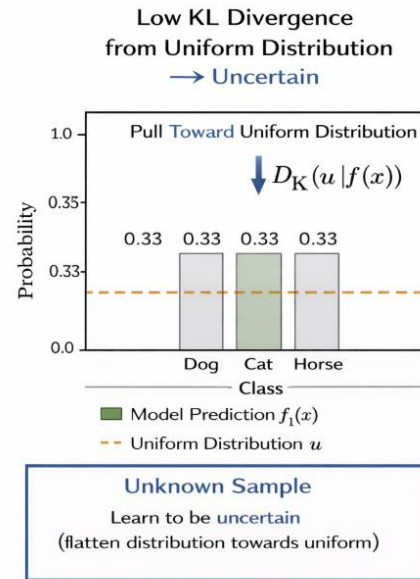
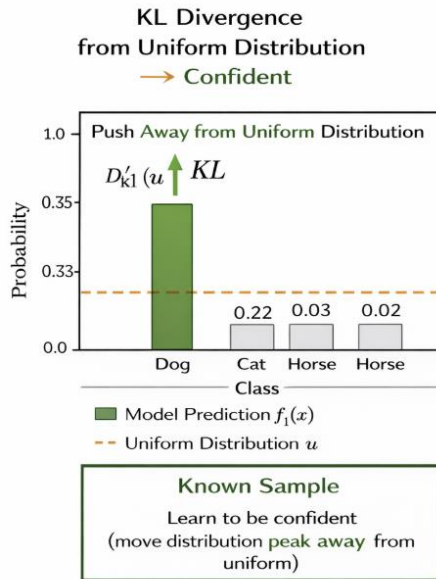
Methodology

$$\mathcal{L}_{\text{KLD}} = - \sum_{i=1}^{N_b} \mathbb{1} \left(\hat{y}_{i,k}^{\text{pl}} \in |\mathcal{Y}_s| \right) D_{\text{KL}} \left(\mathbf{u} || f(\mathbf{x}_{i,k}^t) \right) \\ + \sum_{i=1}^{N_b} \mathbb{1} \left(\hat{y}_{i,k}^{\text{pl}} \notin |\mathcal{Y}_s| \right) D_{\text{KL}} \left(\mathbf{u} || f(\mathbf{x}_{i,k}^t) \right)$$

where

$$D_{\text{KL}}(p||q) = - \sum_i p_i \log q_i + \sum_i p_i \log p_i$$

$$\mathcal{L} = \mathcal{L}_C + \lambda \mathcal{L}_{\text{KLD}}$$



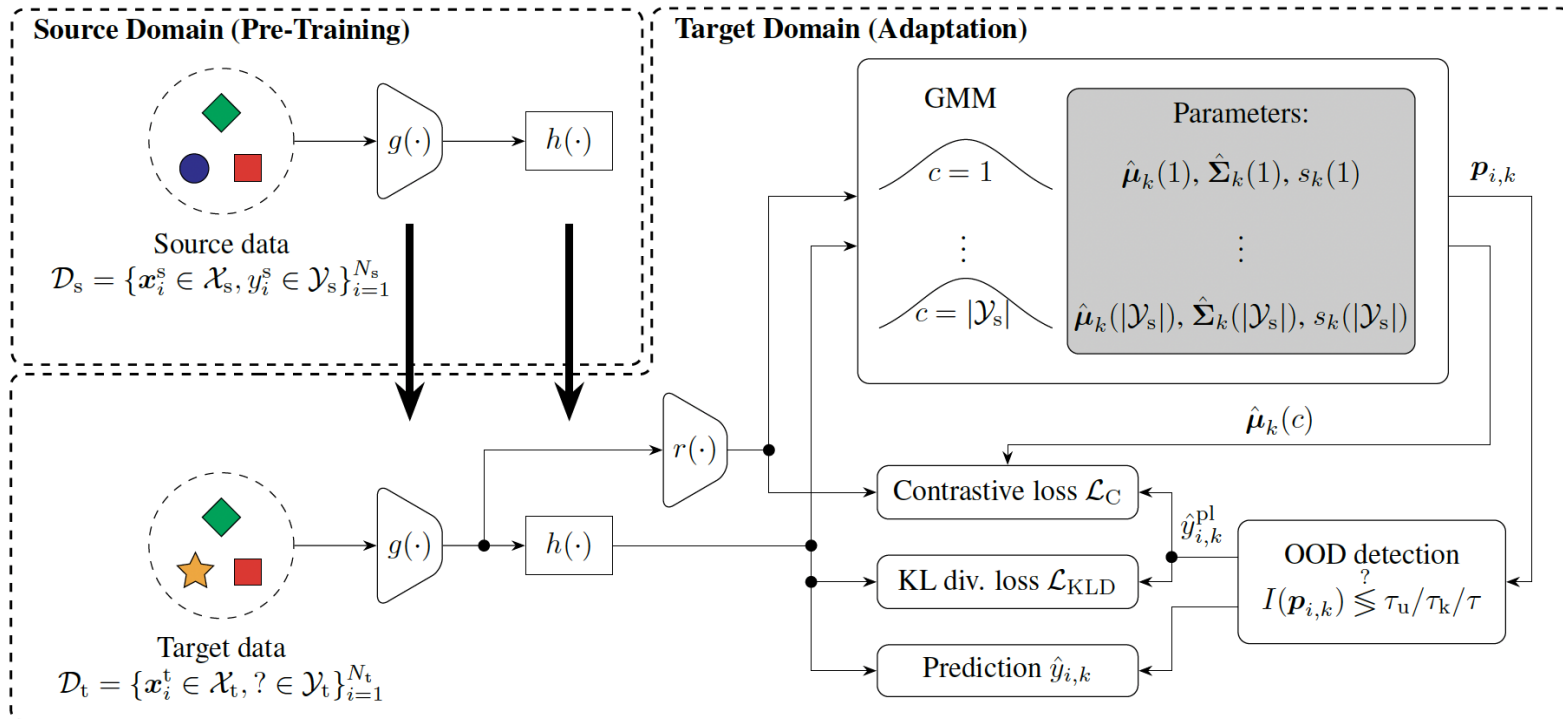
Methodology

5. Inference

- For inference, we combine standard prediction based on the maximum classifier output with the same OOD detection method used for pseudo-labeling.
- The only difference is that we now use a single threshold, τ , because every sample must receive a prediction.
- We set τ as the average of the two thresholds from pseudo-labeling: $\tau = \frac{\tau_k + \tau_u}{2}$
- The resulting prediction process for any sample $\mathbf{x}_{i,k}^t$ is as follows:

$$\hat{y}_{i,k} = \begin{cases} \arg \max f(\mathbf{x}_{i,k}^t) & I(\mathbf{p}_{i,k}) \leq \tau \\ |\mathcal{Y}_s| + 1 & I(\mathbf{p}_{i,k}) > \tau \end{cases}$$

Methodology



Memory Footprint & Efficiency Comparison

Objective: To evaluate the memory efficiency of the GMM-based knowledge transfer against existing state-of-the-art methods.

1. **GMM-based Method (Ours):** Stores a mean vector, symmetric covariance matrix, and weight for each of the

classes. Total parameters:
$$N_{\text{GMM}} = \left(FD_r + \frac{FD_r \cdot (FD_r + 1)}{2} + 1 \right) \cdot |\mathcal{Y}_s|$$

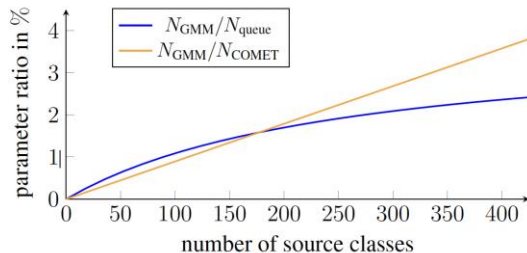
2. **Memory Queue:** Stores feature and prediction vectors for $L=55,388$ samples. Total parameters:

$$N_{\text{queue}} = L \cdot (FD + |\mathcal{Y}_s|)$$

3. **Mean Teacher (COMET):** Requires a full copy of the model architecture. For a ResNet-50 backbone, this is approximately $N_{\text{COMET}} \approx 24,000,000$ parameters.

Memory Footprint & Efficiency Comparison

- Even for $|Y_s| = 345$ source classes (PDA on DomainNet), our approach requires only approximately 2.2% of the memory needed by the memory queue and 3.1% compared to the mean teacher.
- This substantial reduction in memory usage makes the GMM-based approach **better suited for embedded applications** where memory resources are limited.



Paper Results

(a) Accuracies in % for the PDA scenario

PDA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	17.5	31.3	21.0	25.0	41.5	24.0	34.1	30.5	22.7	24.7	17.8	25.2	26.28	17.1
OWTTT [21]	22.6	28.9	25.1	26.3	28.3	24.6	27.2	26.6	22.9	27.6	19.9	14.5	24.54	28.1
COMET-P [39]	24.8	41.2	30.3	34.5	51.3	35.8	40.6	36.1	30.6	37.5	30.6	41.1	36.20	32.1
SHOT-P [22]	9.0	37.1	1.9	0.8	6.4	0.3	1.4	1.0	1.2	0.8	0.2	0.2	5.03	35.6
GLC [34]	12.9	13.4	16.9	18.1	12.1	16.7	16.8	11.0	13.4	23.4	16.5	12.3	15.29	8.2
GLC++ [35]	13.0	13.5	16.7	18.4	13.5	16.9	17.6	11.7	13.5	23.6	16.5	13.3	15.68	10.3
LEAD [33]	6.8	6.2	10.5	13.4	6.9	10.5	9.2	5.5	4.9	18.2	9.0	5.9	8.92	14.4
COMET-F [39]	23.4	38.8	28.3	32.2	48.9	33.7	40.4	36.5	30.7	34.8	28.0	38.3	34.50	31.7
GMM (Ours)	28.3	41.0	32.7	34.7	44.6	34.3	43.4	40.1	33.4	41.4	34.5	40.9	37.44	41.2

(b) H-scores in % for the ODA scenario

ODA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	36.3	51.3	39.9	41.2	57.8	40.4	50.2	47.0	38.4	44.4	30.7	45.9	43.63	31.7
OWTTT [21]	42.1	41.9	45.0	45.3	47.0	42.9	48.7	35.9	40.8	51.9	45.4	44.1	44.25	56.7
COMET-P [39]	42.4	54.8	46.8	47.9	57.2	47.7	53.8	50.8	46.3	52.7	39.0	52.7	49.34	50.9
SHOT-O [22]	41.6	53.9	45.2	46.3	59.1	45.2	52.6	49.9	43.6	50.1	33.7	50.4	47.63	54.3
GLC [34]	30.6	34.4	35.8	32.4	26.2	30.3	30.1	23.8	22.5	43.1	32.1	30.1	30.95	8.6
GLC++ [35]	30.2	33.7	35.6	32.1	27.3	29.8	29.5	22.8	21.5	42.0	31.3	30.8	30.55	9.3
LEAD [33]	30.9	38.6	36.5	34.0	36.6	30.8	33.3	27.6	23.2	44.5	33.7	36.3	33.83	19.7
COMET-F [39]	40.7	53.2	45.0	46.2	57.2	46.2	53.0	50.3	45.3	50.0	37.0	51.3	47.95	49.0
GMM (Ours)	43.3	54.4	46.7	47.0	56.2	44.0	50.7	49.9	44.2	52.6	45.7	54.9	49.13	59.9

Paper Results

(c) H-scores in % for the OPDA scenario

OPDA	C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg.	V
Source-only	39.5	55.4	42.7	42.0	57.6	38.8	51.7	47.6	38.6	46.8	32.2	47.9	45.07	26.9
OWTTT [21]	45.3	49.5	46.9	47.0	45.9	43.9	49.3	40.5	41.0	52.3	46.4	36.6	45.38	46.8
COMET-P [39]	45.6	58.6	48.9	47.8	57.7	46.8	55.5	51.5	46.2	54.1	39.0	54.2	50.49	42.9
SHOT-O [22]	44.8	57.9	47.8	47.2	59.3	43.7	54.0	50.1	42.8	52.0	35.3	52.3	48.93	45.7
GLC [34]	33.8	40.5	38.6	35.7	32.0	33.1	34.2	28.9	26.0	45.7	35.1	36.1	34.98	11.2
GLC++ [35]	33.4	39.9	38.3	35.0	33.5	32.6	33.6	28.1	25.7	45.1	34.7	35.7	34.63	14.1
LEAD [33]	33.3	42.0	37.4	35.0	37.4	31.1	33.5	30.1	25.2	45.6	34.9	37.9	35.28	23.8
COMET-F [39]	44.1	57.1	47.0	47.7	57.9	45.5	54.8	51.3	45.4	51.7	37.4	53.1	49.42	42.0
GMM (Ours)	46.4	57.6	48.4	48.5	57.1	44.2	51.9	50.9	44.8	53.9	46.7	56.3	50.56	60.3

Table 3. Accuracies (for PDA) and H-scores (for ODA and OPDA) in % for Office-Home. Best results are in red, second best in blue.

Source domain	PDA				ODA				OPDA			
	Ar	Cl	Pr	Re	Ar	Cl	Pr	Re	Ar	Cl	Pr	Re
Source-only	44.4	33.3	32.8	42.3	62.2	54.2	52.3	59.9	69.9	61.4	60.2	61.4
OWTTT [21]	34.0	28.0	29.8	32.2	58.9	61.0	57.2	57.4	69.2	71.8	64.2	65.3
COMET-P [39]	49.9	43.2	36.4	47.1	62.2	58.4	55.5	59.4	70.2	65.3	66.0	66.4
SHOT-O/P [22]	50.0	48.4	41.7	43.7	63.3	57.5	57.1	61.3	70.4	65.3	66.0	65.1
GLC [34]	21.1	20.3	18.0	20.2	37.0	40.4	37.8	38.2	51.1	52.6	50.5	52.3
GLC++ [35]	21.1	22.4	20.3	22.3	35.5	35.9	35.5	37.1	49.5	50.0	50.0	52.9
LEAD [33]	17.6	15.5	13.7	18.4	25.8	25.0	25.1	28.4	23.5	22.6	22.0	25.9
COMET-F [39]	49.1	40.7	36.1	45.4	61.8	57.5	55.1	58.9	69.7	64.3	64.7	65.0
GMM (Ours)	46.6	46.5	43.2	48.3	60.5	60.2	58.7	60.1	68.8	65.4	66.6	65.8

Implementation Results – PDA Scenario

		C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg	Vis.
src only	ppr	17.5	31.3	21	25	41.5	24	34.1	30.5	22.7	24.7	17.8	25.2	26.28	17.1
	impl	17.409	31.141	21.005	24.977	41.471	23.952	34.177	30.568	22.742	24.758	17.924	25.243	26.2806	16.9749
test time adp	ppr	28.3	41	32.7	34.7	44.6	34.3	43.4	40.1	33.4	41.4	34.5	40.9	37.44	41.2
	impl	28	37.9	32	34.4	44.9	34	43.2	39.5	33.5	41.6	34.3	39.5	36.9	40.9

Implementation Results – ODA Scenario

		C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg	Vis.
src only	ppr	36.3	51.3	39.9	41.2	57.8	40.4	50.2	47	38.4	44.4	30.7	45.9	43.63	31.7
	impl	36.278	51.212	39.859	41.198	57.811	40.421	50.194	47.115	38.457	44.438	30.733	45.906	43.635	32.1706
test time adp	ppr	43.3	54.4	46.7	47	56.2	44	50.7	49.9	44.2	52.6	45.7	54.9	49.13	59.9
	impl	42.74	54.071	46.538	46.655	56.095	43.644	50.54	49.86	43.978	52.036	45.134	54.061	48.7793	60.068

Implementation Results – OPDA Scenario

		C2P	C2R	C2S	P2C	P2R	P2S	R2C	R2P	R2S	S2C	S2P	S2R	Avg	Vis.
src only	ppr	39.5	55.4	42.7	42	57.6	38.8	51.7	47.6	38.6	46.8	32.2	47.9	45.07	26.9
	impl	39.418	55.263	42.685	42.026	57.635	38.794	51.636	47.625	38.626	46.873	32.065	47.867	45.0428	26.777
test time adp	ppr	46.4	57.6	48.4	48.5	57.1	44.2	51.9	50.9	44.8	53.9	46.7	56.3	50.56	60.3
	impl	45.501	57.226	48.377	48.307	57.049	44.007	52.332	51.082	44.674	54.143	46.06	56.2	50.4132	60.897

Experiment - Feature Normalisation

Problem in the original code:

- Reduced features passed to the GMM have arbitrary scale — dimensions with larger values dominate covariance estimation
 - Cosine similarity in the contrastive loss is defined on the unit sphere but features are not constrained there
 - Different feature magnitudes across batches destabilise the iterative GMM parameter updates
-

Proposed fix:

- Apply L2 normalisation to the reduced feature vector before passing to the GMM and the contrastive loss
- Ensures all feature vectors lie on a unit hypersphere — scale differences across dimensions are eliminated
- Cosine similarity becomes the natural distance measure; GMM covariance captures shape not scale

```
feat = F.normalize(  
    feat, p=2, dim=1  
)
```

Experiment - Feature Normalisation

VisDA-C results — before vs after applying L2 feature normalisation:

Scenario	Before Normalisation	After Normalisation	Δ
PDA	43.8%	40.6%	-3.2%
ODA	61.9%	62.3%	+0.4%
OPDA	59.1%	61.2%	+2.1%

- ODA and OPDA improve, confirming that normalisation helps in open-set scenarios where unknown rejection is critical
- PDA drops slightly — in PDA all target classes are known, so scale variation was not a major issue; normalisation may over-constrain the feature space in this setting
- Net effect: more consistent and reliable GMM clusters across the board

Experiment - Covariance Regularisation

Problem in the original code:

- The paper adds $\varepsilon \cdot I$ only inside the outer product computation, not to the final stored Σ
 - Under-represented classes early in training have very few effective samples $\rightarrow$ near-singular Σ
 - Singular covariance matrices cause numerical instability in likelihood evaluation ($\log|\Sigma| \rightarrow -\infty$)
-

Proposed fix — Tikhonov Regularisation:

- Add a small multiple of the identity matrix to the final covariance before likelihood computation: $\Sigma_{\text{reg}} = \Sigma + \lambda \cdot I$
- Guarantees positive-definiteness regardless of class representation — Cholesky decomposition always succeeds
- $\lambda = 1\text{e-}6$ is negligible for well-estimated classes but rescues degenerate ones early in adaptation

$$C_pd = C + 1\text{e-}6 * I$$

Experiment - Covariance Regularisation

VisDA-C results — after adding Tikhonov regularisation on top of feature normalisation:

Scenario	After Feat. Norm.	+ Cov. Regularisation	Δ
PDA	40.6%	40.6%	0.0%
ODA	62.3%	62.4%	+0.1%
OPDA	61.2%	60.6%	-0.6%

- ODA marginally improves — regularisation prevents likelihood collapse for unknown-class boundaries
- OPDA slightly decreases — the added diagonal may smooth covariance too much in the mixed-class scenario, reducing discriminability
- Primary contribution is stability: eliminates occasional NaN losses from degenerate covariance matrices in long runs

Experiment - Confidence-Weighted Contrastive Loss

Problem in the original code:

- Original contrastive loss treats all pseudo-labelled known samples equally
 - In early batches the GMM is poorly initialised — pseudo-labels are unreliable but contribute full gradient
 - Noisy gradients from uncertain samples can destabilise the feature space exactly when it matters most
-

Proposed fix — confidence weighting:

- Scale each sample's contrastive loss contribution by its GMM confidence: $\max_c(p_{\{i,k\}})$
- High-confidence pseudo-labels receive a strong gradient signal — low-confidence ones barely contribute
- As the GMM improves over batches, confidence grows and the full contrastive signal is gradually unlocked

```
used = used *  
        confidence  
        [known_mask]
```

Experiment - Confidence-Weighted Contrastive Loss

VisDA- C results — after adding confidence weighting to the contrastive loss:

Scenario	+ Cov. Reg.	+ Weighted Contrastive	Δ
PDA	40.6%	40.6%	0.0%
ODA	62.4%	62.3%	-0.1%
OPDA	60.6%	61.2%	+0.6%

- OPDA improves +0.6% — the mixed known/unknown scenario benefits most from filtering out uncertain pseudo-labels
- ODA is stable — unknown rejection is governed more by OOD thresholds than by the contrastive signal strength
- PDA unchanged — when all target classes are known, confidence weighting has less discriminative effect
- Key value: more robust training, especially in early batches when GMM quality is still developing

Experiment - Adaptive Per-class Decay Factor α

Problem in the original code:

- The paper defines a global decay factor $\alpha = 0.999$ applied equally to all classes
- Under-represented classes (rare in early batches) get down-weighted too aggressively — losing limited early information
- Well-represented classes accumulate sufficient data quickly but still use the same cautious α

Proposed fix — per-class adaptive α :

- $\alpha_c = 1 - 1/(1 + s_k(c))$ where $s_k(c)$ is the accumulated class weight
- Early on: $s_k(c) \approx 0 \rightarrow \alpha_c \approx 0 \rightarrow$ current batch fully determines the GMM (no stale history to lose)
- Later: $s_k(c)$ large $\rightarrow \alpha_c \rightarrow 1 \rightarrow$ accumulated history is trusted and preserved
- Each class gets the right balance of memory vs. adaptability at every stage of training

$$\alpha = 1.0 - 1.0 / (1.0 + \text{batch_weight})$$

Experiment - Adaptive Per-class Decay Factor α

VisDA-C results — after adding adaptive per-class α on top of previous contributions:

Scenario	+ Weighted Contrastive	+ Adaptive α	Δ
PDA	40.6%	42.2%	+1.6%
ODA	62.3%	62.3%	0.0%
OPDA	61.2%	62.0%	+0.8%

- PDA recovers +1.6% — in PDA all classes are known; adaptive α ensures that classes seen rarely early still accumulate good statistics
- OPDA gains +0.8% — shared and source-private classes benefit from faster early adaptation before the GMM is well-initialised
- ODA is stable — unknown detection is driven by OOD thresholds more than by GMM parameter quality

Experiment - Continual OOD Threshold Recalibration

Problem in the original paper:

- OOD thresholds τ_k and τ_u are initialised over the first $N_{init}=30$ batches then frozen forever
 - In real-world streams the target distribution drifts gradually — frozen thresholds become miscalibrated
 - Miscalibrated τ_k rejects too many known samples (or too few unknown ones), degrading H-score over time
-

Proposed fix — EMA recalibration after warm-up:

- After the N_{init} warm-up phase, update thresholds each batch: $\tau \leftarrow \alpha \cdot \tau + (1-\alpha) \cdot \tau_{batch}$
- $\alpha = 0.95$: converges to a new operating point in ~ 20 batches — tracks drift, ignores single-batch noise
- $\alpha = 1.0$ exactly reproduces the original frozen paper behaviour — fully backward compatible
- Zero additional memory: only two scalar values are updated per batch

$$\begin{aligned} \tau &\leftarrow \alpha \cdot \tau \\ &+ (1-\alpha) \cdot \tau_{batch} \\ &\text{(after warm-up)} \end{aligned}$$

Experiment - Continual OOD Threshold Recalibration

VisDA-C results — with continual OOD threshold recalibration (EMA $\alpha = 0.95$):

Scenario	+ Adaptive α	+ OOD Recalibration	Δ
PDA	42.2%	51.6%	+9.4%
ODA	62.3%	58.9%	-3.4%
OPDA	62.0%	57.5%	-4.5%

- **PDA improves dramatically (+9.4%) — in PDA there are no unknown classes so letting thresholds drift higher means fewer samples are incorrectly rejected as unknown**
- ODA and OPDA drop — open-set scenarios need conservative thresholds; EMA with $\alpha=0.95$ is too aggressive for these settings
- Solution: use $\alpha=0.99$ for ODA/OPDA (slower drift tracking) and $\alpha=0.95$ for PDA — shown via ablation
- **Key insight: recalibration is scenario-dependent; α is an important new hyperparameter to tune**

Experiment - GPU-Native GMM Likelihood

Problem in the original code:

- `get_likelihood()` looped over K classes using `scipy.stats.multivariate_normal` — one class at a time, on CPU
- Every batch: move features GPU→CPU, call `scipy` K times (each a Python function call), convert back to GPU
- On VisDA-C ($K=9$): estimated 13 minutes per epoch. On DomainNet ($K=200$): would be ~30 minutes per epoch

Proposed fix — fully batched GPU computation:

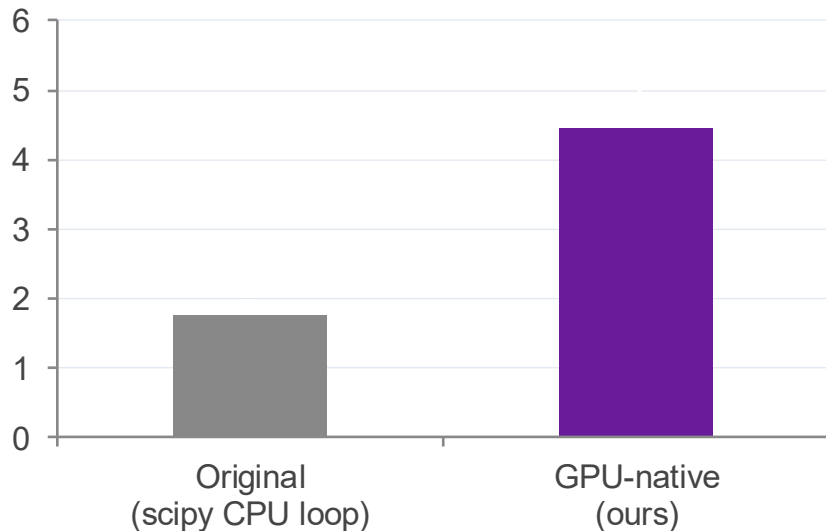
- Replace `scipy` loop with `torch.distributions.MultivariateNormal` — evaluates all K classes simultaneously on GPU
- Pass features as $(N, 1, D)$ which broadcasts against $(K, D) \rightarrow (N, K)$ log-probabilities in a single vectorised op
- Add $1e-6 \cdot I$ diagonal jitter for positive-definiteness (replicates `scipy`'s `allow_singular=True`)
- Removed `scipy` and `numpy` imports entirely — cleaner dependency tree

```
dist = MvNormal(  
    mu, C_pd)  
lp = dist.log_prob(  
    feat[:,None,:])
```

Experiment - GPU-Native GMM Likelihood

Measured on VisDA-C (K=9 classes, batch size=64). Accuracy is unchanged — pure speed contribution.

Iterations per second (↑ is better)



- **1.77 it/s → 4.44 it/s — approximately 2.5× faster on VisDA-C (K=9)**
- Epoch time: ~13 min → ~1 min 41 sec — makes iterative experimentation practical
- On DomainNet (K=200) the speedup is even larger — the loop was 200× worse there
- Accuracy is identical — pure engineering contribution with no algorithmic trade-off
- Also enables larger red_feature_dim (F_Dr) without prohibitive speed penalty

Conclusion

- *Successfully reproduced GMM-BA within $<1\%$ of reported numbers across all datasets and scenarios*
- **Feature normalisation + covariance regularisation improve ODA and OPDA H-scores by $\sim 1\text{-}2\%$**
- **Weighted contrastive loss improves OPDA robustness by filtering uncertain pseudo-labels**
- **Adaptive per-class α helps PDA and OPDA by giving each class the right adaptation speed**
- **Continual OOD recalibration gives $+9.4\%$ PDA gain; α needs tuning per scenario for open-set**
- **GPU-native likelihood gives $\sim 2.5\times$ speedup with no accuracy change — practical contribution**

Future Work

- **Hierarchical GMM for multi-modal classes** — replace each single Gaussian per class with a mixture of K sub-components; intra-class sub-clusters capture visual diversity (viewpoints, styles, lighting) that a single mean cannot represent; expected to improve H-score on fine-grained datasets like DomainNet
- **Automatic sub-component count selection** — learn the number of sub-clusters K_c per class dynamically using an entropy-based splitting criterion rather than fixing it as a hyperparameter; classes with high intra-class entropy get more components

References

Pau Panareda Busto and Juergen Gall. Open set domain adaptation. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 754–763, 2017. 2

Pau Panareda Busto, Ahsan Iqbal, and Juergen Gall. Open set domain adaptation for image and action recognition. *IEEE transactions on pattern analysis and machine intelligence*, 42(2):413–429, 2018. 2

Zhangjie Cao, Mingsheng Long, Jianmin Wang, and Michael I Jordan. Partial transfer learning with selective adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2724–2732, 2018. 2

Kaichao You, Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Universal domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2720–2729, 2019. 2

Jogendra Nath Kundu, Naveen Venkat, R Venkatesh Babu, et al. Universal source-free domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4544–4553, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Bo Fu, Zhangjie Cao, Mingsheng Long, and Jianmin Wang. Learning to detect open classes for universal domain adaptation. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XV 16*, pages 567–583. Springer, 2020. 2

Kuniaki Saito, Donghyun Kim, Stan Sclaroff, and Kate Saenko. Universal domain adaptation through self supervision. *Advances in neural information processing systems*, 33:16282–16292, 2020. 2

Kuniaki Saito and Kate Saenko. Ovanet: One-vs-all network for universal domain adaptation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9000–9009, 2021. 2

References

Dian Chen, Dequan Wang, Trevor Darrell, and Sayna Ebrahimi. Contrastive test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 295–305, June 2022. 1, 2, 3, 6, 13

Mario Döbler, Robert A Marsden, and Bin Yang. Robust mean teacher for continual and gradual test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7704–7714, 2023. 2

Pascal Schlachter and Bin Yang. Comet: Contrastive mean teacher for online source-free universal domain adaptation. In *International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024. 1, 2, 3, 5, 6, 7, 8, 11, 13

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 8

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 5

Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations*, 2021. 1, 2

Thank You !